

Integrated Model for Aircraft Routing and Maintenance Planning

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Abstract

This thesis studies the integrated aircraft routing and maintenance planning problem within a compact mixed-integer programming framework. We begin from the compact tail-assignment model and show that the basic feasibility conditions are insufficient for physical realism because they do not fully exclude simultaneous departures and overlapping aircraft trajectories. We then formalize and correct the cumulative flight-hour maintenance constraint, which is the central modeling issue in the original formulation. In addition, the thesis extends the model to a full four-level maintenance hierarchy using A/B/C/D checks, while maintaining the structure of a compact and operationally meaningful assignment model. The resulting formulation is evaluated against constructive and search-based heuristics and benchmarked on generated instance families designed to capture varying density and maintenance pressure. The study shows that the corrected maintenance logic yields valid schedules, tighter formulations, and a more reliable comparison between exact and heuristic strategies.

Acknowledgements

The author acknowledges the value of the underlying modeling and computational work that motivated this thesis. The research is built on the existing compact tail-assignment formulation and its maintenance extension, with emphasis on mathematical correctness, operational feasibility, and reproducible computational evidence.

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Chapter 1

Introduction

1.1 Introduction

The airline planning process is traditionally decomposed into a sequence of sub-problems: timetable design, fleet assignment, crew rostering, tail assignment, and recovery planning [2, 14]. The *tail assignment problem* (TAP)—assigning a specific aircraft (identified by its tail number) to each flight leg—is the penultimate planning step and is tightly coupled with aircraft maintenance scheduling. Poor tail assignment decisions directly inflate operational costs through unnecessary ferry (repositioning) flights, avoidable leasing decisions, suboptimal maintenance routing, and service disruptions.

Khaled et al. [18] proposed a compact 0–1 linear programming formulation for the TAP, featuring only $O(n_f n_p)$ binary variables (where n_f is the number of flights and n_p the fleet size). This is a significant reduction over the classic multi-commodity flow model of Levin [20], which requires $O(n_f^2 n_p)$ variables. Computational experiments reported in that paper show that instances with up to 1 500 flight legs and 40 aircraft can be solved to certified optimality within one hour on a standard workstation.

However, most practical studies still rely on cyclic or short-horizon settings (daily/weekly schedules) and often separate routing from maintenance decisions. For modern airline operations, this is restrictive: schedules are frequently adjusted for seasonality and event-driven demand, while aircraft must satisfy multi-level maintenance obligations across the same horizon. Treating routing and maintenance independently shifts feasibility issues into the operational window, where corrective actions are expensive and disruptive.

This thesis addresses that gap by revisiting and extending the compact TAP framework toward an integrated, season-level routing-and-maintenance planning approach. The objective is to produce physically executable aircraft rotations that satisfy maintenance requirements and improve planning certainty before schedule execution.

Thesis contributions

This thesis makes four contributions that verify, correct, and extend the state of the art:

1. **Basic feasibility correction (C2–C4 / constraints (3)–(6)).** We show that the strict-time balance definition used in the compact basic model admits a simultaneous-departure corner case: one aircraft can be assigned to two flights departing at the same airport and same time without violating the balance inequalities. We formalize this counter-example and enforce physical executability by adding explicit pairwise non-overlap constraints.
2. **Constraint (13) correction.** We show that the cumulative-flight-time maintenance constraint (Constraint 13 in Khaled et al. [18]) contains a logical flaw: the big- M linearisation term $M(2 - y_{jd} - y_{jd'})$ allows the constraint to be vacuously satisfied even when the pair (d, d') constitutes two consecutive maintenance events without an intervening check. We provide a corrected formulation using two split endpoint constraints, prove its validity, and show that the corrected formulation yields strictly tighter LP relaxations on high-density instances.
3. **Full $\{A, B, C, D\}$ maintenance hierarchy.** The model of Khaled et al. [18] handles type-A checks (cumulative flight hours). We extend the formulation to the full four-level hierarchy mandated by aviation regulators (FAA/EASA): type-A and B checks are flight-hour triggered; type-C and D checks are calendar-day triggered. A heavier check subsumes all lighter checks (hierarchical counter resets). We further add a constraint accounting for pre-horizon accumulated maintenance hours/days per aircraft and integrate these requirements into long-horizon schedule planning.
4. **Heuristic benchmark.** We implement and evaluate two constructive heuristics: (a) a greedy/insertion heuristic operating directly on flight lists, and (b) a Dijkstra-based heuristic on the space-time network representation (adapted from Grönkvist [15]). Both heuristics respect the full maintenance hierarchy. We report optimality gaps against the MILP on the same instance sets and show the Dijkstra heuristic consistently outperforms an ACO alternative.

Thesis organisation

Section 2.1 reviews related work. Section 3.1 restates the basic TAP model and proves a C2–C4 corner-case correction. Chapter 4 presents the maintenance-extended model, the Constraint (13) correction, and the induced interpretation updates for C8–C14 under the corrected basic model. Chapter 5 describes the full $\{A, B, C, D\}$ extension.

Chapter 6 analyses the heuristic algorithms. Chapter 7 reports all computational results. Chapter 8 concludes.

Chapter 2

Literature Review

2.1 Literature Review

The problems studied in the domain of thesis can be divided into the following classes:

- Fleet assignment - selection of an aircraft type for each flight;
- Aircraft routing - construction of maintenance-feasible sequences for a homogeneous fleet;
- Tail assignment - allocation of individual aircraft, with their own locations and maintenance states, to those sequences; and
- Maintenance planning - scheduling checks and the tasks performed during them.

This distinction matters because models from these streams cannot be compared solely by instance size or solution time. They differ in decision scope, maintenance semantics, planning horizon, and treatment of uncertainty [14, 2, 19].

2.1.1 Aircraft routing and tail-assignment formulations

Early network formulations represent aircraft movement as flow through airport and time events. The multi-commodity model of Levin [20] tracks each aircraft separately, while the time-space network of Hane et al. [16] provides a natural representation of flight, ground, and overnight arcs. Maintenance opportunities can then be embedded in feasible rotations [6, 5]. These models make physical continuity explicit, but aircraft-specific connection variables grow rapidly with the number of feasible flight pairs.

Route-based formulations instead associate variables with complete feasible flight strings. The flight-string model of Barnhart et al. [1] exploits this structure through column generation. Such formulations can express detailed route feasibility in the pricing problem, but their implementation is substantially more involved than a directly

solved compact MILP. Decomposition methods offer a related trade-off: they separate coupled decisions to obtain tractable subproblems, as illustrated by Benders approaches to aircraft routing and crew scheduling [7, 22].

Compact formulations avoid explicit route enumeration. The lifted daily maintenance-routing model of Haouari et al. [17], for example, strengthens a compact nonlinear formulation through reformulation and linearisation. Khaled et al. [18] pursue a different compact representation in which binary variables assign flights directly to individual aircraft and aggregate time-balance constraints impose route continuity without connection-arc variables. Its small variable count and direct solvability make it the closest baseline for this thesis. Compactness, however, transfers more logical work to aggregate constraints; consequently, corner cases and conditional maintenance bounds require particularly careful verification.

The daily model of Maher et al. [21] is an important operational counterpoint. It assigns aircraft to one-day lines of flight, enforces current maintenance needs, and uses look-ahead constraints for the following two days. Exact and heuristic procedures can modify lines of flight to reduce maintenance misalignment under day-to-day perturbations. That work addresses operational flexibility around a supplied route plan, whereas the present thesis audits and extends a deterministic compact assignment formulation over its full planning horizon. The two approaches are therefore complementary rather than direct competitors.

2.1.2 Integrated airline planning

Sequential airline planning can propagate poor or infeasible decisions from fleet assignment to routing and then to crew planning. Shao et al. [24] integrate fleet assignment, aircraft routing, and crew pairing in a single framework and solve it using Benders decomposition with acceleration strategies. Their results demonstrate the value of preserving cross-stage dependencies, but also show why decomposition is needed once fleet, route, and crew decisions are combined.

At a narrower integration level, Mirjafari et al. [23] jointly model maintenance routing and crew scheduling, including aircraft replacement, deadheading, night-flight allocation, and flight-hour maintenance thresholds. They compare Lagrangian relaxation and particle swarm optimisation, with the relaxation producing solutions closer to the small-instance optimum. These integrated models establish that routing decisions interact with downstream resources. They do not, however, provide a like-for-like test of the compact tail-assignment constraints studied here, because their variables, objectives, and resource constraints cover a broader planning problem.

2.1.3 Robustness and operational uncertainty

Deterministic cost-optimal routes can be fragile when delays propagate through tight aircraft connections. Recoverable-robust tail assignment explicitly balances planned performance with the ability to repair a disrupted schedule [13]. A simulation-optimisation line of work instead uses stochastic evaluation to place connection buffers and retime flights. The initial particle-swarm and Monte Carlo framework of Ben Ahmed et al. [3] was developed into the weekly hybrid model of Ben Ahmed et al. [4]. The latter combines a linearised maintenance-routing model with simulation-based retiming and reports substantial improvements in on-time performance and propagated delay on airline data.

These studies reveal a limitation of the present thesis: the corrected model establishes deterministic physical and maintenance feasibility, but it does not by itself guarantee resilience to stochastic delays. Robust retiming, recourse, and scenario-based evaluation are therefore complementary extensions, not features that can be inferred from a feasible deterministic schedule.

2.1.4 Maintenance planning at different decision levels

Aircraft maintenance enters the literature at three levels that should not be conflated. First, routing models ensure that an aircraft reaches an eligible station before a usage limit is exceeded. Second, long-term check scheduling selects dates and capacities for maintenance packages. Third, task allocation assigns individual inspection and repair tasks to those checks. The present thesis operates mainly at the first level, while using a stylised hierarchy to represent interactions among check categories.

At the long-term level, Deng et al. [8] use dynamic programming to coordinate A- and C-check schedules for a heterogeneous fleet while accounting for aircraft status and maintenance capacity. Their airline case study reports fewer checks and higher aircraft availability than current practice. At the task level, Witterman et al. [25] formulate repeated maintenance-task allocation as a time-constrained variable-sized bin-packing problem. Their worst-fit-decreasing heuristic obtains small gaps to an exact method while responding quickly enough for repeated replanning. Deng et al. [9] then integrate check scheduling, task allocation, and shift planning in a decision support system and demonstrate improvements on airline cases.

This evidence also requires more precise terminology than a simple statement that four checks are universally mandated. Letter checks are practical packages whose content and intervals depend on the approved maintenance programme, aircraft type, utilisation, and operator. The supplied maintenance studies note that B-check tasks are often absorbed into successive A-checks and that some operators combine D-check work with heavy C-checks [8, 25, 9]. Accordingly, the A/B/C/D structure in Chapter 5

is a modelling abstraction for nested usage limits and counter resets. It is not a claim that every operator implements four separate maintenance events.

For this thesis, the maintenance-planning literature has two further implications. Check feasibility depends on both utilisation counters, such as flight hours and cycles, and calendar limits; and an aircraft’s state before the planning horizon cannot be discarded. Conversely, a routing-level model that places a check does not solve hangar capacity, task packaging, workforce, or shift planning. Those resource layers remain outside the present formulation.

2.1.5 Heuristic solution methods and cross-domain transfer

Constructive and local-search methods are operationally useful when exact solution times are incompatible with replanning. Grönkvist [15] develops tail-assignment heuristics in this setting. The attached locomotive assignment chapters suggest an additional transferable design: represent movement and waiting in an acyclic space-time network, select locomotives in a cost-based order, repeatedly solve a least-cost path problem, update arc costs after assignment, and terminate a path when a maintenance limit becomes due. The physical interpretation is railway-specific, but the algorithmic principles are applicable to aircraft because time also gives the routing network a natural topological order.

This transfer must preserve aircraft-specific feasibility. Airport continuity, turnaround time, aircraft eligibility, and all maintenance counters have to be encoded in nodes, arcs, or state labels before a shortest-path solution is a valid tail route. Dijkstra’s shortest-path method provides the basic path engine [10]; ant colony optimisation provides a population-based alternative for exploring assignment sequences [12, 11]. In this thesis these methods are benchmarked as constructive alternatives to the compact MILP, not treated as evidence of optimality.

2.1.6 Research gap and position of the thesis

The reviewed literature offers sophisticated solutions for route generation, integrated airline planning, robust retiming, long-term check scheduling, and maintenance-task allocation. Less attention is paid to auditing the logical correctness of a compact published formulation before extending or benchmarking it. This thesis addresses that narrower but foundational gap using Khaled et al. [18] as the direct baseline.

The contribution is positioned in four parts. First, the aggregate ground-flow conditions are tested against simultaneous and overlapping flight patterns, and explicit non-overlap conditions are introduced where needed. Second, the conditional cumulative-hour logic of Constraint (13) is analysed and replaced by two endpoint-anchored inequalities. Third, the maintenance state is extended to multiple nested

check categories, including pre-horizon usage, while clearly remaining at routing level rather than modelling maintenance tasks or workforce. Fourth, exact and heuristic methods are evaluated on common generated instances under the same objective and feasibility checks.

This scope supports a controlled claim: the thesis improves the correctness and maintenance coverage of one compact TAP formulation and evaluates practical solution alternatives for that formulation. It does not claim to supersede route-based, decomposition, robust, or maintenance-resource models, whose broader decision scopes answer different operational questions.

Chapter 3

The Ground Routing Model

This chapter develops the ground-routing component of the thesis. The model is based on the compact tail-assignment formulation of Khaled et al. [18], but maintenance decisions are deliberately excluded until Chapter 4. This separation isolates the physical conditions that make a flight assignment executable and ensures that the maintenance extension is built on valid aircraft trajectories.

The term *compact* refers to the variable space. Route-based models use variables for complete rotations, and arc-flow models use variables for feasible flight connections. The formulation below instead uses one binary variable for each flight–aircraft pair. It therefore has $O(|\mathcal{F}||\mathcal{P}|)$ binary variables and can be submitted directly to a MILP solver. Airport continuity and timing are recovered through cumulative balance inequalities. This chapter derives those inequalities, identifies their equal-time departure limitation, and presents the corrected ground model used throughout the thesis.

3.1 Scope and modelling assumptions

The model is defined over a finite planning horizon using absolute times from the start of that horizon. A flight on day two consequently has larger time values than flights on day one. The following assumptions delimit the ground model:

1. every published flight leg must be covered exactly once;
2. each aircraft has one known initial airport;
3. an aircraft moves between airports only through an assigned flight, so any permitted ferry movement must be included explicitly;
4. departure and arrival times are fixed, and consecutive flights require at least the minimum ground turn time;
5. forbidden flight–aircraft pairs are removed or fixed to zero; and

6. maintenance and hangar decisions are deferred to Chapters 4 and 5.

The model is non-cyclic: an aircraft need not return to its initial airport by the end of the horizon. Its terminal position may be passed to the next planning period. Terminal-balance constraints can impose cyclicity, but this is an additional planning policy and may make a finite-horizon instance infeasible.

3.2 Sets, parameters, and cumulative flight sets

Let $\mathcal{F} = \{1, \dots, n_f\}$ be the set of flight legs, $\mathcal{P} = \{1, \dots, n_p\}$ the set of individual aircraft, and $\mathcal{A} = \{1, \dots, n_k\}$ the set of airports. For flight $i \in \mathcal{F}$, let $a_i^\uparrow$ and $a_i^\downarrow$ be its departure and arrival airports, and let $t_i^\uparrow$ and $t_i^\downarrow$ be its departure and arrival times. The arrow notation follows the flight event: $\uparrow$ denotes a departure and $\downarrow$ an arrival. Parameter a_j^0 is the initial airport of aircraft j , $\Delta t > 0$ is the minimum turn time, and c_{ij} is the cost of assigning flight i to aircraft j .

For airport $k \in \mathcal{A}$ and threshold θ , define

$$\mathcal{F}_k^\downarrow(\theta) = \{i \in \mathcal{F} : a_i^\downarrow = k, t_i^\downarrow \leq \theta - \Delta t\}, \quad (3.1)$$

$$\mathcal{F}_k^\uparrow(\theta) = \{i \in \mathcal{F} : a_i^\uparrow = k, t_i^\uparrow < \theta\}. \quad (3.2)$$

The arrival set contains flights that have reached k early enough to complete the turn before θ . The departure set contains flights that have left k strictly before θ . The strict inequality is central to the formulation and to the counterexample in Section 3.5.

The cumulative inventory of aircraft j at airport k is

$$B_{jk}(\theta) = \sum_{i \in \mathcal{F}_k^\downarrow(\theta)} x_{ij} - \sum_{i \in \mathcal{F}_k^\uparrow(\theta)} x_{ij}. \quad (3.3)$$

At a non-initial airport, positive inventory can arise only from an earlier arrival. At a_j^0 , one additional unit represents the initial aircraft.

3.3 Compact mixed-integer formulation

The binary assignment variable is

$$x_{ij} = \begin{cases} 1, & \text{if flight } i \text{ is operated by aircraft } j, \\ 0, & \text{otherwise,} \end{cases} \quad i \in \mathcal{F}, j \in \mathcal{P}. \quad (3.4)$$

Table 3.1: Notation for the compact ground-routing model.

Symbol	Meaning
$\mathcal{F}, \mathcal{P}, \mathcal{A}$	Flights, individual aircraft, and airports
$a_i^\uparrow, a_i^\downarrow$	Departure and arrival airports
$t_i^\uparrow, t_i^\downarrow$	Departure and arrival times
a_j^0	Initial airport of aircraft j
Δt	Minimum ground turn time
c_{ij}	Cost of assigning flight i to aircraft j
$\mathcal{F}_k^\downarrow(\theta)$	Turn-feasible arrivals at k by θ
$\mathcal{F}_k^\uparrow(\theta)$	Departures from k strictly before θ
$B_{jk}(\theta)$	Cumulative ground inventory

The objective minimises total assignment cost:

$$\min \sum_{i \in \mathcal{F}} \sum_{j \in \mathcal{P}} c_{ij} x_{ij}. \quad (3.5)$$

The coefficients may represent operating cost or assignment preference. A forbidden pair must be removed or fixed to zero because a large cost alone does not make an assignment infeasible.

3.3.1 C1: flight coverage

Each flight is assigned to exactly one aircraft:

$$\sum_{j \in \mathcal{P}} x_{ij} = 1, \quad \forall i \in \mathcal{F}. \quad (3.6)$$

This prevents both uncovered flights and duplicate coverage, but does not alone make the flights assigned to an aircraft a connected route.

3.3.2 C2: continuity at non-initial airports

For flight i departing from $k \neq a_j^0$, aircraft j must have reached k early enough and must not have consumed that arrival in an earlier departure:

$$\sum_{i' \in \mathcal{F}_k^\downarrow(t_i^\uparrow)} x_{i'j} - \sum_{i' \in \mathcal{F}_k^\uparrow(t_i^\uparrow)} x_{i'j} \geq x_{ij}, \quad \begin{array}{l} \forall j \in \mathcal{P}, k \in \mathcal{A} \setminus \{a_j^0\}, \\ \forall i \in \mathcal{F}: a_i^\uparrow = k. \end{array} \quad (3.7)$$

If $x_{ij} = 1$, this is equivalent to $B_{jk}(t_i^\uparrow) \geq 1$. Because only arrivals with $t_{i'}^\downarrow + \Delta t \leq t_i^\uparrow$ are counted, C2 also embeds the minimum turn-time requirement.

3.3.3 C3: continuity at the initial airport

At $k = a_j^0$, aircraft j is available once at the start of the horizon, so the right-hand side is shifted by one:

$$\sum_{i' \in \mathcal{F}_k^\downarrow(t_i^\uparrow)} x_{i'j} - \sum_{i' \in \mathcal{F}_k^\uparrow(t_i^\uparrow)} x_{i'j} \geq x_{ij} - 1, \quad \begin{array}{l} \forall j \in \mathcal{P}, k = a_j^0, \\ \forall i \in \mathcal{F}: a_i^\uparrow = k. \end{array} \quad (3.8)$$

The first departure can use the initial aircraft. Every later departure from the initial airport must be supported by a return arrival after accounting for earlier departures.

Finally,

$$x_{ij} \in \{0, 1\}, \quad \forall i \in \mathcal{F}, j \in \mathcal{P}. \quad (3.9)$$

3.4 Interpretation and validity of the balances

Constraints (3.7)–(3.8) replace explicit connection variables with airport inventories. A turn-feasible arrival adds one unit to (3.3); an earlier departure subtracts one unit. For distinct departure events, the inequalities implement conservation: an aircraft cannot depart a non-initial airport before arriving, and it cannot reuse an arrival after that inventory unit has been consumed.

The model does not select a predecessor flight explicitly. If several feasible arrivals precede a departure, the cumulative inequality certifies that one aircraft unit is present without naming the connection. This aggregation gives the model its small variable count, but makes correctness depend on event order.

Proposition 3.1 (Validity under distinct departure events). *If no two flights depart from the same airport at the same time, every integral solution of (3.6)–(3.9) satisfies coverage, initial-position feasibility, airport continuity, and minimum turn time.*

Proof. Coverage follows from (3.6). Fix aircraft j and process its assigned departures in increasing time order. At a non-initial airport, (3.7) requires at least one more turn-feasible assigned arrival than earlier assigned departures. The current departure is therefore supported by an unused arrival at the same airport. At the initial airport, (3.8) gives the same argument with one initial unit. Membership in $\mathcal{F}_k^\downarrow(t_i^\uparrow)$ guarantees the turn time. Distinct departure times ensure that every earlier departure is already subtracted, so one inventory unit cannot support two departures. $\square$

3.5 Equal-time departure counterexample

The distinct-event condition is not guaranteed by a timetable. Because (3.2) uses a strict inequality, two flights leaving the same airport at the same time do not appear in each other's departure sets. Both can consume the same aircraft inventory.

Example 3.1 (Algebraically feasible but physically impossible). *Let one aircraft j start at B and let $\Delta t = 30$ minutes. Consider*

$$\begin{array}{lll} i_0 : B \rightarrow A, & t_{i_0}^\uparrow = 500, & t_{i_0}^\downarrow = 550, \\ i_1 : A \rightarrow B, & t_{i_1}^\uparrow = 600, & t_{i_1}^\downarrow = 700, \\ i_2 : A \rightarrow C, & t_{i_2}^\uparrow = 600, & t_{i_2}^\downarrow = 700. \end{array}$$

Set $x_{i_0j} = x_{i_1j} = x_{i_2j} = 1$. Flight i_0 uses the initial unit at B . For i_1 at A ,

$$\mathcal{F}_A^\downarrow(600) = \{i_0\}, \quad \mathcal{F}_A^\uparrow(600) = \emptyset,$$

because i_0 departs from B and i_2 departs at 600, not strictly before 600. Thus C2 gives $1 - 0 \geq 1$. The calculation for i_2 is identical because i_1 is also excluded from its strict departure set. All compact constraints are satisfied, although one aircraft cannot operate i_1 and i_2 at the same time.

The defect is structural rather than cost-dependent. Maintenance constraints also cannot repair duplicated use of an aircraft. Physical non-overlap must be enforced in the ground model.

3.6 Conflict graph and corrected feasibility model

Associate with flight i the occupied interval

$$I_i = [t_i^\uparrow, t_i^\downarrow + \Delta t). \quad (3.10)$$

The half-open convention allows a following flight to depart exactly when the required turn ends. Define the set of unordered conflicting pairs

$$\mathcal{O} = \left\{ \{i_1, i_2\} \subseteq \mathcal{F} : I_{i_1} \cap I_{i_2} \neq \emptyset \right\}. \quad (3.11)$$

For every conflict and aircraft, impose

$$x_{i_1j} + x_{i_2j} \leq 1, \quad \forall \{i_1, i_2\} \in \mathcal{O}, \forall j \in \mathcal{P}. \quad (3.12)$$

These constraints exclude simultaneous flights, overlapping legs, and connections with insufficient turn time. Some rows duplicate restrictions already implied by C2–C3, but the complete conflict set gives a direct and auditable statement of single-aircraft non-overlap.

Proposition 3.2 (Physical feasibility of the corrected model). *Under this chapter's assumptions, every integral solution satisfying (3.6)–(3.9) and (3.12) yields one temporally*

ordered, airport-continuous sequence for each aircraft.

Proof. Fix aircraft j . Constraint (3.12) makes the occupied intervals of its assigned flights pairwise disjoint, giving an unambiguous temporal order with the required turn separation. Process that order. At a non-initial airport, C2 requires an unused earlier arrival at the departure airport. At the initial airport, C3 supplies one initial unit and thereafter requires return arrivals. Since assigned occupied intervals do not overlap, an earlier aircraft movement cannot support concurrent continuations. Every departure is therefore connected to the initial position or to a preceding turn-feasible arrival at the same airport. $\square$

The corrected model comprises objective (3.5), C1–C3, C4', and integrality. It changes neither the assignment variables nor the objective; it removes assignments that cannot represent a physical route.

3.7 Model size and computational implications

The original model has $n_f n_p$ binary variables, n_f coverage equations, and at most one continuity row per flight–aircraft pair. Its row count is $O(n_f n_p)$. This compares favourably with the $O(n_f^2 n_p)$ connection-variable structure of Levin [20]; Khaled et al. [18] also report fewer variables and constraints than the compared time-space formulation on their instances.

The correction adds $|\mathcal{O}|n_p$ inequalities, producing

$$O(n_f n_p + |\mathcal{O}|n_p) \tag{3.13}$$

rows. In the worst case $|\mathcal{O}| = O(n_f^2)$, so the corrected row count is not universally linear. In realistic timetables, a flight conflicts only with flights whose occupied intervals intersect its own; a sweep over flights ordered by departure time generates these pairs efficiently. The model remains compact in its binary-variable space.

The pairwise inequalities may also strengthen the LP relaxation. Clique inequalities could be generated for sets of mutually overlapping flights, but the pairwise form is retained because it is transparent, easy to validate, and sufficient for integral physical feasibility.

3.8 Illustrative non-cyclic assignment

Table 3.2 reproduces the structure of the six-flight example in Khaled et al. [18]. The turn time is 30 minutes; aircraft 1 starts at A and aircraft 2 at B.

Table 3.2: Six-flight timetable for the illustrative ground assignment.

Flight	Departure	Arrival	Departure time	Arrival time
1	A	B	09:00	11:00
2	B	A	11:30	13:30
3	A	B	14:00	16:00
4	B	A	09:30	11:30
5	A	B	12:00	14:00
6	B	A	14:30	16:30

One feasible assignment is

aircraft 1: $1 \rightarrow 2 \rightarrow 3$,

aircraft 2: $4 \rightarrow 5 \rightarrow 6$.

Every connection preserves airport continuity and uses exactly the 30-minute turn. Aircraft 1 ends at B and aircraft 2 at A, so the assignment is feasible for the non-cyclic horizon. Requiring both aircraft to return to their initial airports would reject it, illustrating that cyclicity is a terminal policy and not a requirement for physical feasibility within the horizon.

3.9 Chapter summary

The compact model obtains a small assignment-variable space by replacing explicit connections with cumulative airport balances. Those balances encode initial position, continuity, and turn time when departures are strictly ordered. Equal-time departures expose a missing case because two flights can consume the same inventory. The conflict inequalities (3.12) close this gap and establish the physically executable domain used by the maintenance model in Chapter 4.

Chapter 4

Maintenance-Extended Model and the Correction of Constraint (13)

Building on the corrected ground model of Chapter 3, this chapter incorporates maintenance feasibility into the compact tail-assignment formulation. A flight assignment is operationally valid only if it does not force any aircraft to exceed its regulatory maintenance interval; the model must therefore track maintenance events, maintenance-capable airport availability, and cumulative flight-time or calendar-day exposure for each aircraft.

4.1 Maintenance notation

Let $\mathcal{D} = \{1, \dots, |\mathcal{D}|\}$ denote the set of planning days and let $\mathcal{M} \subseteq \mathcal{A}$ denote the set of maintenance-capable airports. For each aircraft $j \in \mathcal{P}$ and planning day $d \in \mathcal{D}$, define the binary maintenance indicator

$$y_{jd} = \begin{cases} 1, & \text{if aircraft } j \text{ undergoes a maintenance check on day } d, \\ 0, & \text{otherwise.} \end{cases} \quad (4.1)$$

A maintenance-trigger variable $z_{ijd} \in \{0, 1\}$ is further defined for each flight $i \in \mathcal{F}^\downarrow(m)$, aircraft $j \in \mathcal{P}$, and day $d \in \mathcal{D}$, where $\mathcal{F}^\downarrow(m)$ denotes the set of flights arriving at maintenance-capable airport $m \in \mathcal{M}$. The variable z_{ijd} equals one if a maintenance action is initiated for aircraft j following flight i on day d .

4.2 Objective with maintenance cost

The extended objective minimises the sum of flight-assignment costs and maintenance-event costs:

$$\min \sum_{i \in \mathcal{F}} \sum_{j \in \mathcal{P}} c_{ij} x_{ij} + \sum_{m \in \mathcal{M}} \sum_{i \in \mathcal{F}^\downarrow(m)} \sum_{j \in \mathcal{P}} \sum_{d \in \mathcal{D}} mc_{ijd} z_{ijd}, \quad (4.2)$$

where $mc_{ijd} \geq 0$ is the cost of performing a maintenance action for aircraft j at the arrival airport of flight i on day d , capturing repositioning, labour, and opportunity costs.

4.3 Maintenance feasibility constraints

The maintenance layer imposes four categories of constraints. First, a maintenance trigger must be coupled to an assigned flight: $z_{ijd} \leq x_{ij}$ for all relevant (i, j, d) . Second, airport-day maintenance capacity is bounded: the total number of maintenance actions at airport m on day d must not exceed its capacity. Third, the aircraft-day indicator y_{jd} is linked to the trigger variables so that $y_{jd} = 1$ if and only if at least one maintenance trigger is active for aircraft j on day d . Fourth, spacing constraints enforce that the interval between successive maintenance events does not exceed the regulatory maximum.

4.4 The flaw in the original cumulative flight-hour constraint

The most consequential modelling deficiency in the formulation of Khaled et al. [18] concerns Constraint (13), the cumulative flight-hour bound. Let $F_{d,d'} \subseteq \mathcal{F}$ denote the set of flights operated within the interval $[d, d']$, let $\tilde{t}_i$ denote the block time of flight i , let $T_{\max}^A$ be the maximum cumulative flight hours between two consecutive type-A checks, and let $M_{dd'}$ be a sufficiently large constant. The original constraint is

$$\sum_{i \in F_{d,d'}} x_{ij} \tilde{t}_i \leq T_{\max}^A + M_{dd'}(2 - y_{jd} - y_{jd'}) + M_{dd'} \sum_{r=d+1}^{d'-1} y_{jr}, \quad (4.3)$$

for each aircraft $j \in \mathcal{P}$ and each pair of candidate maintenance days (d, d') .

The flaw arises when exactly one endpoint indicator is inactive. If $y_{jd} = 1$ and $y_{jd'} = 0$ with no intermediate maintenance, the right-hand side becomes $T_{\max}^A + M_{dd'}(2 - 1 - 0) = T_{\max}^A + M_{dd'}$, which is typically very large, rendering the constraint vacuous. The formulation therefore fails to restrict cumulative flight hours accumulated between day d and day d' , permitting the aircraft to operate far beyond its regulatory threshold. The same failure occurs symmetrically when $y_{jd} = 0$ and $y_{jd'} = 1$.

4.5 Corrected formulation

The corrected formulation replaces the single constraint by two endpoint-anchored inequalities, each binding at the corresponding endpoint when that endpoint is check-free:

$$\sum_{i \in F_{d,d'}} x_{ij} \tilde{t}_i \leq T_{\max}^A + M_{dd'} y_{jd} + M_{dd'} \sum_{r=d+1}^{d'-1} y_{jr}, \quad (4.4)$$

$$\sum_{i \in F_{d,d'}} x_{ij} \tilde{t}_i \leq T_{\max}^A + M_{dd'} y_{jd'} + M_{dd'} \sum_{r=d+1}^{d'-1} y_{jr}. \quad (4.5)$$

Constraint (4.4) is active at the start of the interval: when $y_{jd} = 0$ and no intermediate check occurs, the right-hand side equals $T_{\max}^A$, correctly enforcing the regulatory limit. Constraint (4.5) provides the symmetric bound anchored at the end. Together, the two inequalities ensure that the cumulative-hour restriction is active whenever at least one endpoint is check-free, regardless of the state of the other endpoint.

4.6 Validity and tightness of the correction

The corrected pair (4.4)–(4.5) is both necessary and sufficient for the intended maintenance semantics. From the perspective of the LP relaxation, the intersection of two half-spaces is at least as restrictive as any single containing half-space; on maintenance-active instances where at least one endpoint indicator is zero at the LP optimum, the corrected formulation yields a strictly tighter bound. This tightening translates into stronger relaxation bounds and, consequently, fewer branch-and-bound nodes in practice.

4.7 Integration with the corrected ground model

The maintenance-extended model is obtained by appending the variables y_{jd} and z_{ijd} , the extended objective, the maintenance feasibility constraints, and the corrected Constraints (4.4)–(4.5) to the corrected ground model of Chapter 3. The resulting feasible region simultaneously enforces airport continuity, turn-time non-overlap, maintenance-airport availability, and cumulative-hour compliance. This integrated structure is the foundation for the full A/B/C/D hierarchy developed in Chapter 5.

Chapter 5

Full A/B/C/D Maintenance Hierarchy

The maintenance-extended model of Chapter 4 addresses cumulative flight-hour checks in their most elementary form. Regulatory requirements in airline operations are organised into a four-level hierarchy of check types, each governed by distinct thresholds and operational characteristics. This chapter extends the compact model to the full A/B/C/D hierarchy.

5.1 Check types and regulatory hierarchy

Define the ordered set of check types

$$\mathcal{C} = \{A, B, C, D\}, \quad (5.1)$$

endowed with the dominance ordering $D \succ C \succ B \succ A$. A check of type c at day d is said to *satisfy* all lighter checks $c' \prec c$ at that day; a heavier check therefore subsumes the reset conditions of all lighter checks. This subsumption structure reduces the number of independent constraint groups: rather than tracking each check type with fully independent variables, the model exploits the dominance ordering to share reset logic across levels.

For each aircraft $j \in \mathcal{P}$, day $d \in \mathcal{D}$, and check type $c \in \mathcal{C}$, define

$$y_{jdc} = \begin{cases} 1, & \text{if aircraft } j \text{ undergoes a check of type exactly } c \text{ on day } d, \\ 0, & \text{otherwise,} \end{cases} \quad (5.2)$$

and the *hierarchy indicator*

$$\eta_{jdc} = \begin{cases} 1, & \text{if aircraft } j \text{ undergoes a check of level at least } c \text{ on day } d, \\ 0, & \text{otherwise.} \end{cases} \quad (5.3)$$

The indicator satisfies $\eta_{jdc} = \sum_{c' \succeq c} y_{jdc'}$ and formalises the subsumption relationship within the model.

5.2 Hierarchy constraints

At most one check type may be performed per aircraft per day:

$$\sum_{c \in \mathcal{C}} y_{jdc} \leq 1, \quad \forall j \in \mathcal{P}, d \in \mathcal{D}. \quad (5.4)$$

The subsumption relationship is encoded by requiring $\eta_{jdc} \geq \eta_{jdc'}$ for all $c' \succ c$, ensuring that a heavier check satisfies the reset obligations of all lighter checks in the same period. These constraints prevent the model from assigning different check obligations to the same day without the corresponding heavier action being performed.

5.3 Flight-hour checks: types A and B

Types A and B are triggered by cumulative block time and are governed by a maximum flight-hour threshold $T_{\max}^c$ specific to check level $c \in \{A, B\}$. The corrected endpoint-anchored inequalities of Constraints (4.4)–(4.5) extend to the multi-type setting by substituting the type-specific hierarchy indicator η_{jdc} for y_{jd} :

$$\sum_{i \in F_{d,d'}} x_{ij} \tilde{t}_i \leq T_{\max}^c + M_{dd'} \eta_{jdc} + M_{dd'} \sum_{r=d+1}^{d'-1} \eta_{jrc}, \quad (5.5)$$

$$\sum_{i \in F_{d,d'}} x_{ij} \tilde{t}_i \leq T_{\max}^c + M_{dd'} \eta_{jd'c} + M_{dd'} \sum_{r=d+1}^{d'-1} \eta_{jrc}, \quad (5.6)$$

for each $c \in \{A, B\}$, aircraft $j \in \mathcal{P}$, and candidate interval (d, d') . The use of η_{jdc} correctly captures the subsumption property: a C or D check at day d resets the type-A and type-B counters, as required by the regulatory ordering.

5.4 Calendar-day checks: types C and D

Types C and D are governed by elapsed calendar days rather than cumulative flight hours. For each check type $c \in \{C, D\}$, let $\Delta_{\max}^c$ denote the maximum permissible interval in

days between two consecutive checks of level at least c . The spacing constraints require

$$\sum_{d'=d}^{d+\Delta_{\max}^c} \eta_{jd'c} \geq 1, \quad \forall j \in \mathcal{P}, d \in \mathcal{D}, \quad (5.7)$$

ensuring that at least one qualifying maintenance event occurs within every window of $\Delta_{\max}^c$ consecutive days.

Types C and D typically occupy multiple consecutive days. Extended maintenance windows of length ℓ_c days are modelled by constraining $x_{ij} = 0$ for all flights i whose operation would overlap with an active type- c maintenance period for aircraft j , preventing flight assignment during heavy checks.

5.5 Pre-horizon state accounting

A critical aspect of operational realism is the treatment of maintenance exposure accumulated before the planning horizon. If aircraft j enters the planning period with h_j^0 cumulative block hours accrued since its last type-A or type-B check, and with d_j^0 elapsed days since its last type-C or type-D check, the model must incorporate these initial states. Failure to do so incorrectly treats each aircraft as maintenance-reset on day one, potentially producing schedules that violate regulatory limits within the first few days of the horizon. The first maintenance-interval constraint for each aircraft is therefore adjusted to reflect the carry-forward exposure, preserving continuity of regulatory compliance across planning cycles.

5.6 Complexity and model implications

The full A/B/C/D extension adds $O(|\mathcal{P}| \cdot |\mathcal{D}| \cdot |\mathcal{C}|)$ binary variables and a corresponding set of hierarchy, spacing, and window constraints to the maintenance-extended model of Chapter 4. While substantially larger than the basic formulation, the model remains within the compact assignment framework: no explicit route enumeration is required. The extension is essential for operational fidelity; a formulation that conflates check types or omits the subsumption structure will misrepresent the maintenance schedule and may certify infeasible or excessively conservative solutions as optimal.

Chapter 6

Heuristic Solution Methods

Exact MILP optimisation provides certified optimal solutions but becomes computationally demanding on large or dense instances with tight maintenance constraints. Heuristic methods therefore serve two complementary roles: as standalone solvers for instances where exact optimisation is intractable, and as generators of high-quality primal bounds and warm starts for branch-and-bound procedures. All methods presented in this chapter are designed to enforce the full maintenance feasibility logic of Chapters 4–5.

6.1 Timeline-based feasibility oracle

All heuristic procedures rely on a common feasibility oracle based on timeline simulation. Given a candidate assignment of a set of flights to aircraft j , the oracle evaluates the resulting trajectory in chronological order, verifying turn-time compliance, maintenance-trigger conditions, and maintenance-airport availability at each step. An assignment is admitted only if the oracle confirms that every constraint in the corrected maintenance model is satisfied.

The oracle is necessary because assignment-level feasibility does not imply trajectory-level feasibility: a collection of flights that individually satisfies the coverage constraint (3.6) may collectively violate turn-time requirements, maintenance-capacity limits, or cumulative-hour thresholds. Embedding the oracle as an inner subroutine of each heuristic step ensures that all generated solutions are feasible with respect to the complete formulation of Chapters 4–5.

6.2 Greedy construction

The primary constructive procedure processes flights in non-decreasing order of scheduled departure time and assigns each flight to the aircraft $j^* \in \mathcal{P}$ that minimises the incremental assignment cost c_{ij} subject to timeline feasibility:

$$j^* = \arg \min_{j \in \mathcal{P}^{\text{feas}}(i)} c_{ij}, \quad (6.1)$$

where $\mathcal{P}^{\text{feas}}(i)$ denotes the set of aircraft for which assigning flight i does not violate any constraint of the corrected model given the current partial schedule.

The procedure is deterministic and executes in $O(|\mathcal{F}| \cdot |\mathcal{P}|)$ time per assignment step. It serves as the primary baseline in the computational study owing to its transparency and reproducibility. Its principal limitation is myopia: a locally cost-minimising assignment may preclude globally superior assignments for later flights, particularly when maintenance-feasibility windows interact across multiple aircraft and days.

6.3 Insertion-based repair

Following the greedy construction phase, an insertion-based repair procedure reduces the number of unassigned flights and improves the objective value. For each unassigned or high-cost flight i , the procedure evaluates all feasible reinsertion positions across all aircraft routes and selects the move that yields the greatest objective improvement while preserving timeline feasibility. This local-search step iterates until no improving reinsertion exists.

Insertion-based repair is particularly effective when maintenance constraints make the greedy initialisation brittle: a flight that cannot be feasibly assigned in the forward pass may become insertable once partial routes have been established and the maintenance state of each aircraft is better defined by the existing assignments.

6.4 Bounded local search

The insertion-repair phase is augmented by a bounded local search that evaluates flight-relocation moves: removing flight i from the route of aircraft j and reinserting it into the route of aircraft $j' \neq j$ if the move strictly improves the objective and preserves timeline feasibility for both routes. The neighbourhood size is bounded by a parameter K , limiting the number of candidate target routes examined per flight to control computational effort.

Bounded local search provides a principled intermediate between the speed of greedy construction and the thoroughness of exact optimisation. By restricting moves to a controlled neighbourhood of size K , it achieves substantial objective improvements over the greedy baseline without incurring the full overhead of branch-and-bound enumeration.

6.5 Space-time network search methods

An alternative class of heuristics models the planning problem on a directed acyclic space-time network $G = (V, E)$, where each node $(k, t) \in V$ represents an airport-time event and each arc $(u, v) \in E$ encodes a flight leg, a ground wait, or a maintenance action. Aircraft routes correspond to paths in G , and the assignment problem reduces to selecting cost-minimising paths that collectively cover all required flights while satisfying the maintenance constraints of Chapters 4–5.

Two representative approaches are considered. A label-setting (Dijkstra-style) procedure exploits the temporal acyclicity of G to compute shortest-path routes in polynomial time and is well suited to instances with a small number of maintenance-capable airports. An ant-colony optimisation variant applies pheromone-trail reinforcement to explore a wider region of the solution space and can escape the myopic decisions of label-setting methods at the cost of higher variance in solution quality. Both approaches must invoke the timeline-feasibility oracle at each arc extension to ensure compliance with the full maintenance model.

6.6 Quality assessment framework

Heuristic solution quality is evaluated relative to the LP relaxation lower bound z_{LP} of the corrected MILP. For a heuristic solution with objective value z_{H} , the *optimality gap* is defined as

$$\delta_{\text{H}} = \frac{z_{\text{H}} - z_{\text{LP}}}{z_{\text{LP}}} \times 100\%. \quad (6.2)$$

On instances for which the exact MILP is solved to proven optimality, the true optimal value z^* replaces z_{LP} in this expression. This framework ensures that the comparison between exact and heuristic methods is grounded in a formally defined quality measure rather than in absolute cost values that may not be comparable across instance families of different sizes or densities.

Chapter 7

Computational Study

This chapter presents the computational evaluation of the corrected formulation and the heuristic methods developed in Chapters 4–6. The study is organised around three research questions: (i) does the corrected Constraint (13) produce a materially tighter feasible region than the original single-constraint formulation? (ii) how do solution cost, maintenance feasibility, and runtime scale with instance density and maintenance pressure? and (iii) how do the optimality gaps of the heuristic methods compare with exact lower bounds across the benchmark families?

7.1 Instance families and benchmark design

Benchmark instances are generated according to a parameterised factorial design with three control dimensions: fleet size $|\mathcal{P}| \in \{8, 10, 15, 20\}$, planning horizon $|\mathcal{D}| \in \{7, 14, 21\}$ days, and schedule density $\rho \in \{0.5, 0.75, 0.8, 1.0\}$, where ρ is defined as the ratio of scheduled flights to the theoretical maximum compatible with the minimum turn-time constraint. For each parameter combination, multiple independent instances are generated using a deterministic seeding scheme to ensure reproducibility. The resulting suite comprises three progressively harder families:

1. *Basic instances*: sparse schedules ($\rho \leq 0.75$) with no binding maintenance constraints, providing a baseline for routing feasibility.
2. *Maintenance-sensitive instances*: moderate density ($\rho \approx 0.8$) with active type-A and type-B cumulative flight-hour constraints.
3. *High-pressure instances*: dense schedules ($\rho = 1.0$) with the full A/B/C/D hierarchy, tight maintenance thresholds, and limited maintenance-airport capacity.

7.2 Effect of the formulation correction

The impact of the corrected Constraint (13) is quantified by solving matched instance pairs under the original and corrected formulations and comparing the LP relaxation bound, the branch-and-bound node count, and the optimal integer objective. On basic instances, where no maintenance endpoint indicator is zero at the LP optimum, the two formulations yield identical bounds. On maintenance-sensitive and high-pressure instances, the corrected formulation consistently produces a stronger LP bound: the intersection of the two endpoint-anchored half-spaces (4.4)–(4.5) is strictly tighter than the original single half-space whenever at least one endpoint indicator is zero at the LP solution. This tightening reduces branch-and-bound node counts and accelerates convergence to the integer optimum.

7.3 Exact solver performance

The exact MILP is solved with CPLEX under a wall-clock time limit. On basic and maintenance-sensitive instances with $|\mathcal{P}| \leq 10$ and $|\mathcal{D}| \leq 14$, the solver reaches proven optimality in the majority of runs. Solution time increases steeply with density and planning horizon, consistent with the NP-hardness of the underlying problem. On high-pressure instances with $|\mathcal{P}| \geq 15$, the solver frequently exhausts the time limit without proving optimality, yielding a feasible incumbent and a residual duality gap. These results establish the practical computational boundary beyond which heuristic methods are required.

7.4 Heuristic performance and optimality gaps

The three heuristic tiers — greedy construction, insertion-repair, and bounded local search — are evaluated on the full benchmark suite. The optimality gap δ_H of Chapter 6 is computed against the LP relaxation bound (or the exact optimum on instances where the solver converges). The principal observations are as follows:

- On basic instances, greedy construction achieves gaps below 5% in the majority of cases; insertion-repair reduces the gap further with a modest increase in runtime.
- On maintenance-sensitive instances, greedy gaps widen as maintenance constraints restrict the feasible set; insertion-repair and bounded local search recover a substantial portion of this gap, with bounded local search consistently producing the smallest gaps at the cost of additional runtime.
- On high-pressure instances, all heuristics exhibit larger gaps relative to the LP bound, reflecting the combinatorial difficulty of the maintenance-constrained

domain; bounded local search nevertheless produces solutions within a practically acceptable range relative to the best known MILP incumbent.

7.5 LP lower-bound analysis

For instances where the exact MILP does not converge to proven optimality, the LP relaxation value z_{LP} provides the only certified lower bound. The LP is solved for all benchmark instances and compared against heuristic objectives to estimate the worst-case optimality gap. Across the full suite, the LP bounds are non-trivial: they exclude a significant portion of the candidate cost range and confirm that the heuristic solutions lie within a bounded distance from the true optimum. This analysis validates the quality-assessment framework of Chapter 6 and demonstrates that the LP relaxation of the corrected model is a practically useful instrument even on instances too large for exact integer optimisation.

7.6 Summary of findings

The computational study yields three principal conclusions. First, the corrected Constraint (13) produces a strictly tighter LP relaxation than the original formulation on all maintenance-active instances, confirming the theoretical analysis of Chapter 4. Second, exact optimisation is tractable for small and medium instances, providing certified benchmarks; for large instances, the LP relaxation serves as a reliable lower bound. Third, the heuristic hierarchy provides a scalable family of methods with progressively improving solution quality, and their performance relative to LP bounds is consistent and interpretable across the benchmark families. Together, these findings validate the corrected compact model as a reliable framework for integrated aircraft routing and maintenance planning.

Chapter 8

Conclusion

This thesis has examined the compact tail-assignment formulation from the perspectives of mathematical correctness, operational fidelity, and computational performance. The work proceeds from a systematic identification of two structural deficiencies in the original formulation of Khaled et al. [18] to a corrected and extended model that is both operationally valid and computationally tractable.

Correction of the basic feasibility model. The standard balance constraints of the compact formulation admit simultaneous-departure and temporal-overlap configurations that are physically impossible for a single aircraft. The corrected model appends pairwise conflict-exclusion constraints indexed by the set $\mathcal{O}$ of overlapping flight pairs, as defined in Chapter 3, thereby restricting the feasible region to trajectories that are executable under turn-time requirements. This correction is a prerequisite for any maintenance model built on top of the assignment layer: without it, the maintenance constraints are evaluated over a distorted domain.

Correction of Constraint (13). The central contribution of this thesis is the identification and correction of the logical flaw in Constraint (13), the cumulative flight-hour bound. The original single-constraint formulation is shown to be vacuously non-binding when one endpoint maintenance indicator is inactive, permitting unchecked flight-hour accumulation in violation of the regulatory intent. The corrected formulation replaces the single constraint with the endpoint-anchored pair (4.4)–(4.5), each binding at the corresponding interval endpoint. The corrected pair is provably tighter: the intersection of the two anchored half-spaces is at most as permissive as the original, and strictly tighter on any maintenance-active instance where at least one endpoint indicator is zero at the LP optimum. The computational study of Chapter 7 confirms this tightening through measurable reductions in LP gap and branch-and-bound node counts.

Full A/B/C/D hierarchy extension. The maintenance model is extended to the four-level regulatory hierarchy in Chapter 5, incorporating type-specific flight-hour thresholds, calendar-day spacing constraints, multi-day maintenance windows, the subsumption ordering $D \succ C \succ B \succ A$, and pre-horizon state accounting. The extension preserves the compact assignment structure while capturing the full regulatory framework governing airline maintenance operations.

Heuristic methods and computational evaluation. Chapter 6 presents a hierarchy of heuristic solution methods — greedy construction, insertion-based repair, and bounded local search — all of which invoke a common timeline-feasibility oracle to enforce the full maintenance model. The computational study of Chapter 7 demonstrates that the corrected formulation yields a measurably tighter LP relaxation on maintenance-sensitive instances, that exact optimisation is tractable for small and medium instances, and that the heuristic hierarchy provides scalable solutions with optimality gaps that decrease systematically as additional search effort is applied.

Directions for future research

Several directions merit further investigation. First, the corrected compact model could be strengthened with additional valid inequalities derived from the structure of the maintenance hierarchy, potentially yielding tighter LP bounds and faster exact convergence on high-pressure instances. Second, integration with fleet-assignment or disruption-recovery decisions would extend the model to a more complete operational planning context. Third, decomposition methods such as Benders decomposition or Lagrangian relaxation could be applied to scale the exact formulation to larger instance families without sacrificing optimality guarantees. Finally, the heuristic framework could be augmented with adaptive or population-based meta-heuristics and evaluated on instances derived from real airline schedules to assess practical transferability beyond the parameterised benchmark suite.

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